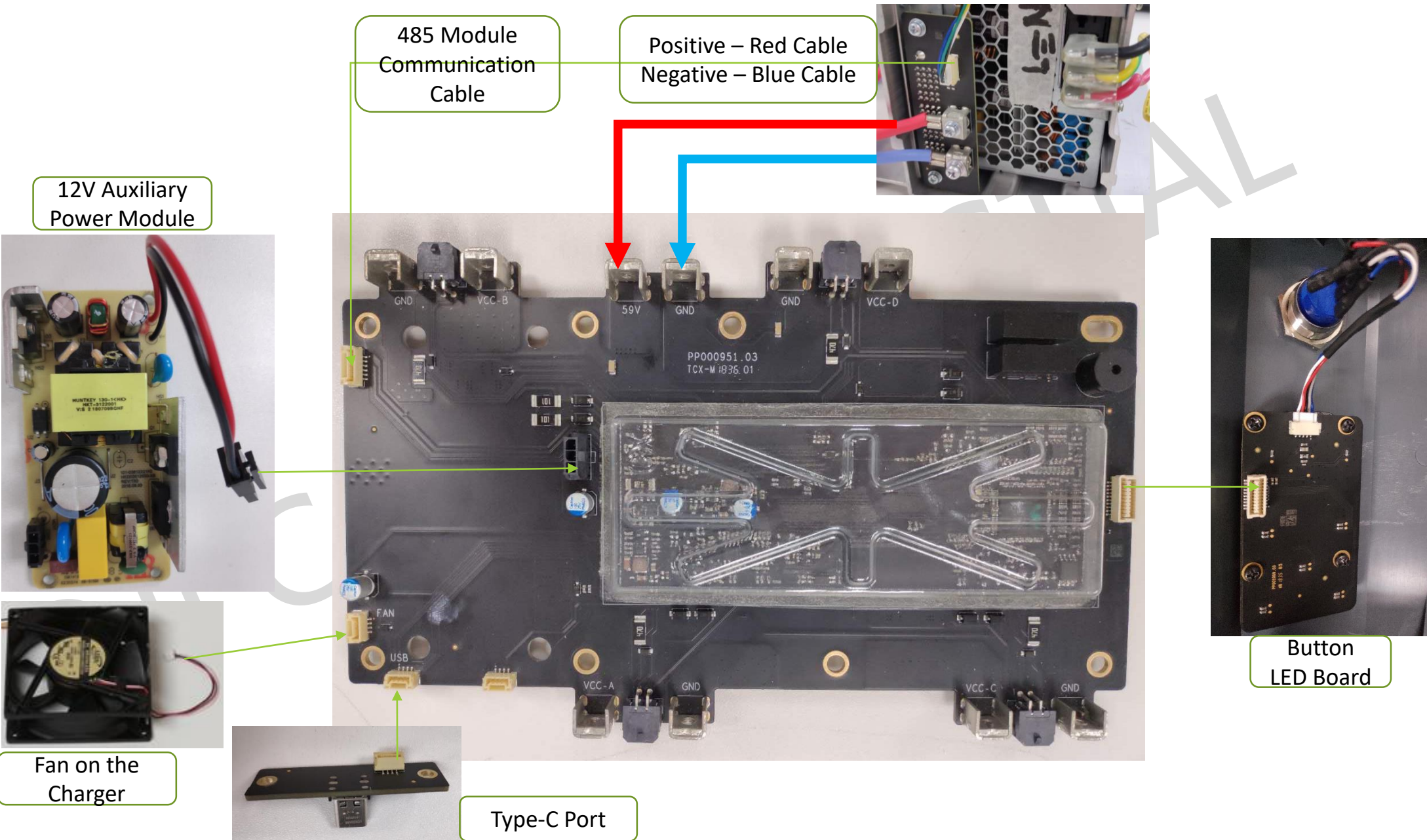




4-Channel Battery Charger Repair Training

4-Channel Battery Charger - Link Diagram



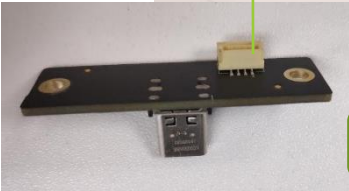
485 Module Communication Cable

Positive – Red Cable
Negative – Blue Cable

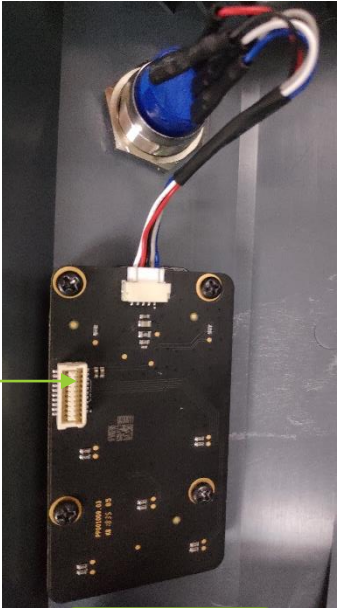
12V Auxiliary Power Module



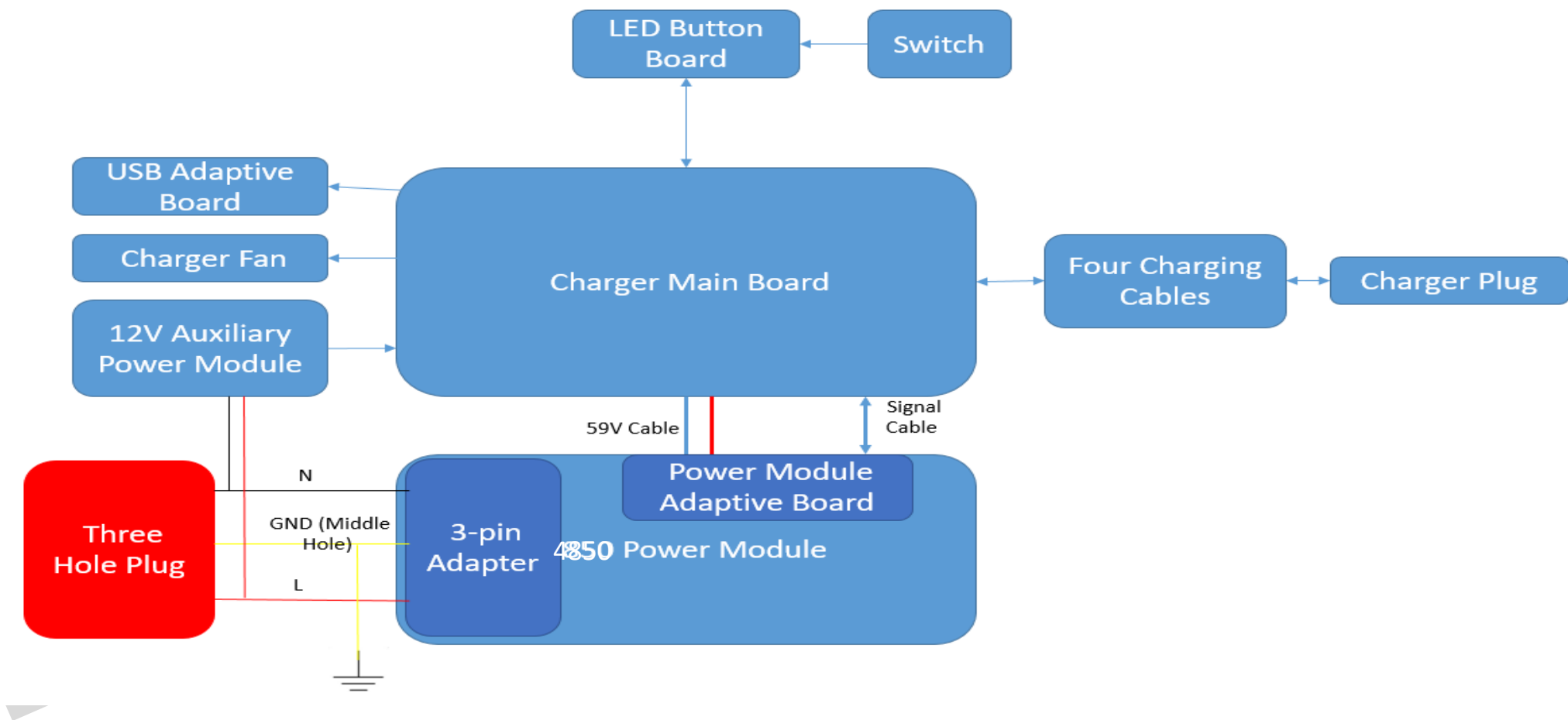
Fan on the Charger

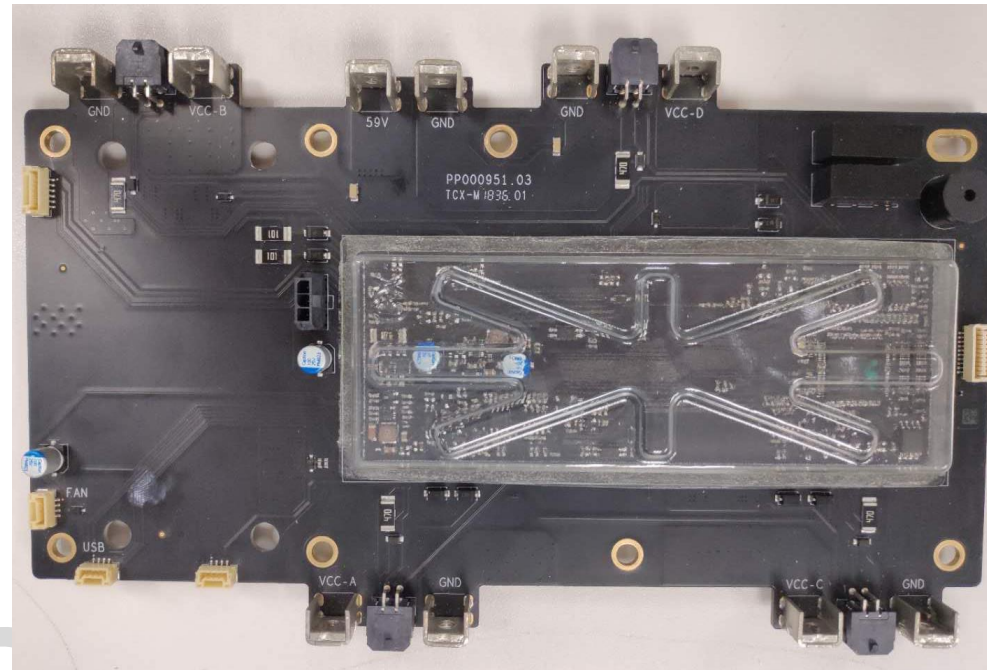


Type-C Port



Button LED Board





Charger management main board: Communicates with the battery; manages and controls the four output ports for charger fast charging, slow charging, and determines which battery is charged first; controls the color of LEDs based on the charging situation and charger status; communicates with the computer via USB port board; exports error logs; connects to the computer for updates; and communicates with the 4850 power module and manages the 4850 power module output.

Auxiliary Power & Fan & USB Port Adaptive Board



12V Auxiliary Power: Transfers electric supply power to the 12V DC power supply and supplies power to the charger management main board.

Fan on the charger: Cools down the charger. Not interchangeable with the fan inside the 4850 module.

USB Port Adaptive Board: Adapts the USB port to the main board. Used to connect the computer for activation and updates, and connects the remote controller via the OTG cable to export the abnormal charging data.

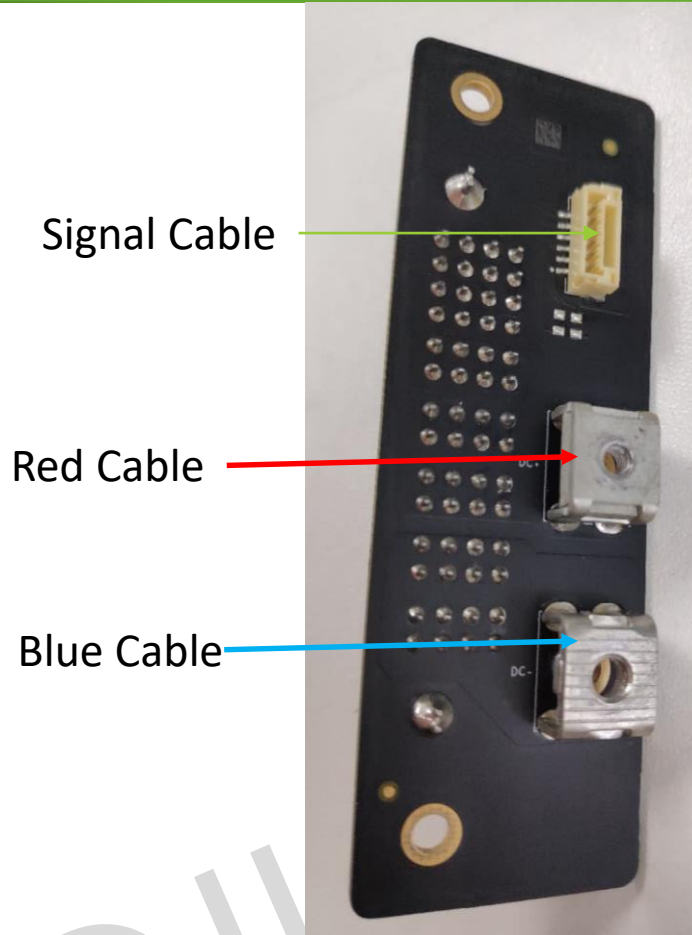
Three-Hole Plug



Fire wire has an "L" mark, called the Live Wire; zero line has an "N" mark, called the Naught Wire. Another cable is the earth wire, which connects the 4850 module on the charger and can be used to ensure safety.

If the cables are connected incorrectly, the product may not work, or may produce a shock hazard.

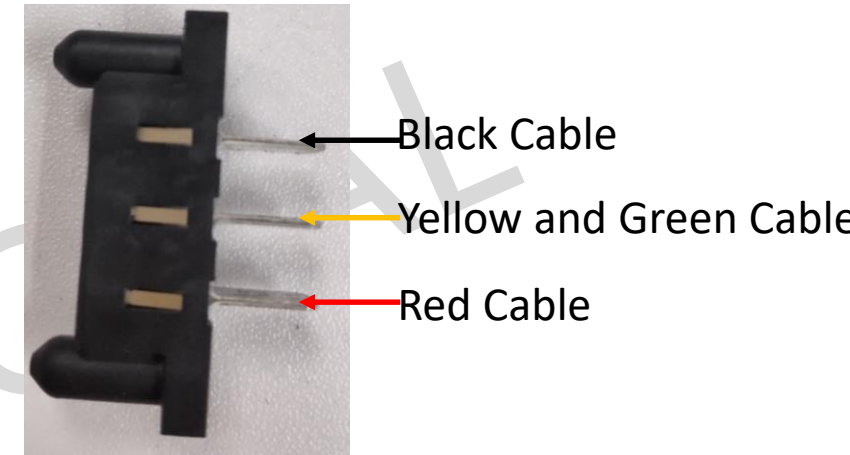
4850 Module



Power Module Adaptive Board



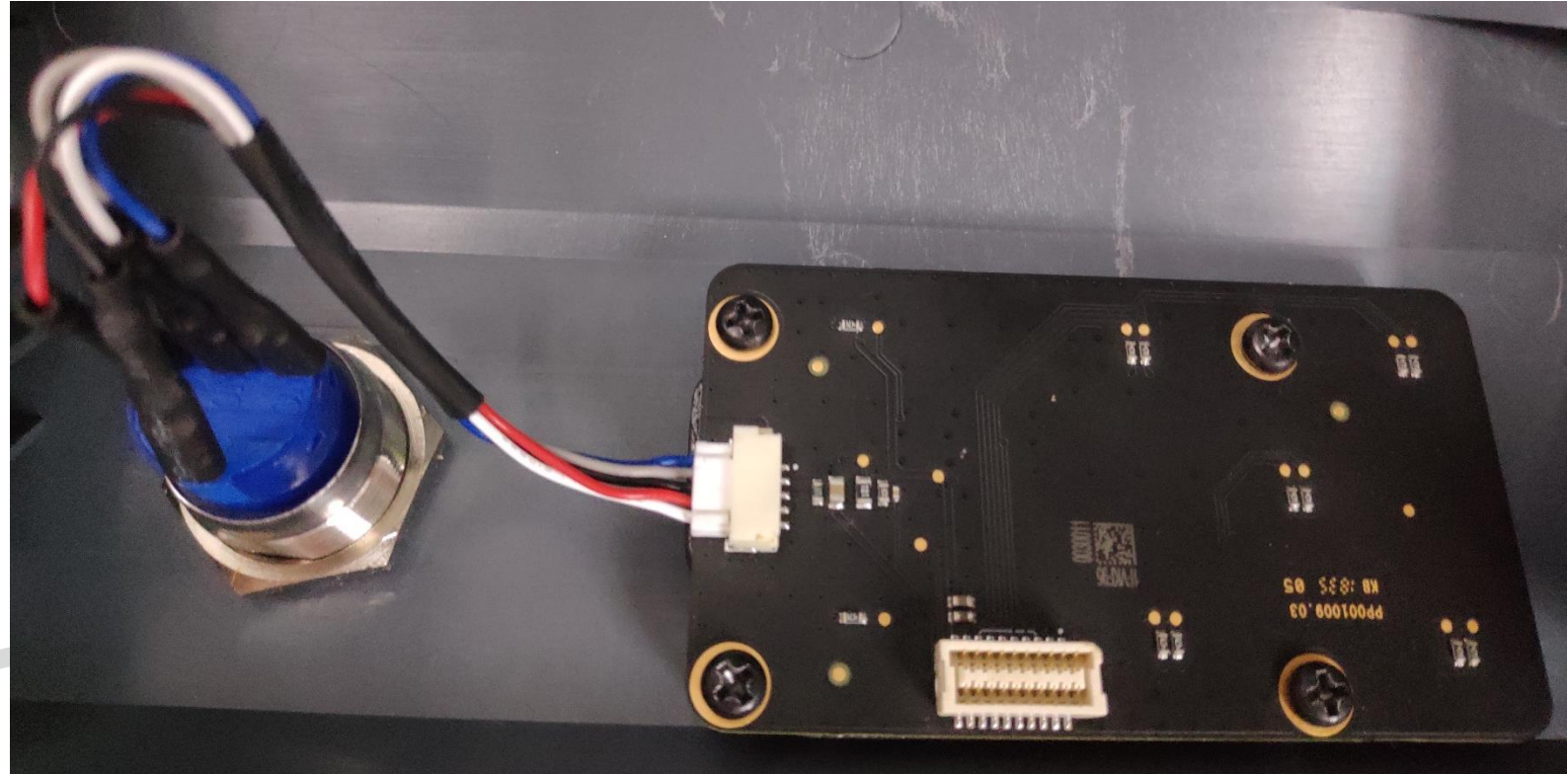
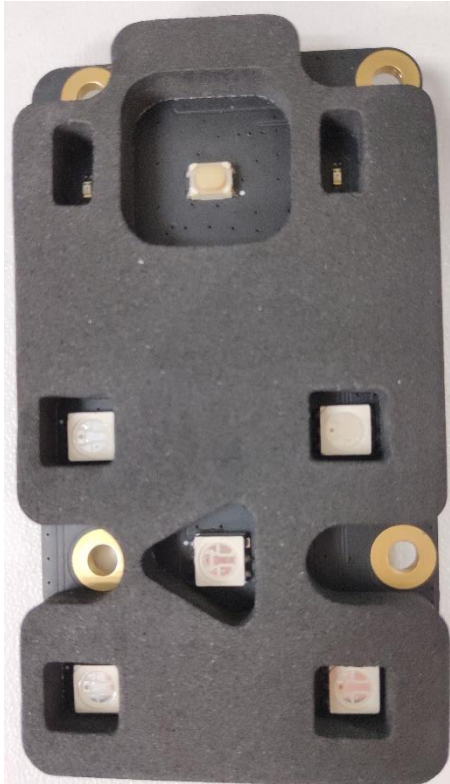
4850 Power Module



3-pin Adapter

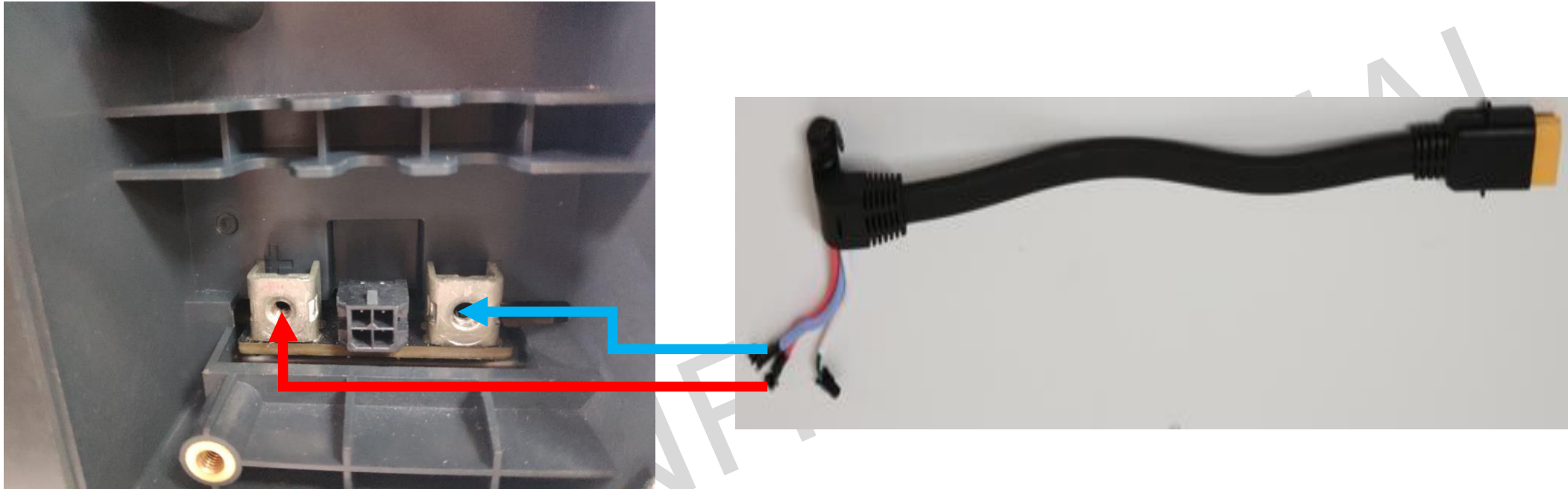
4850 power module: Transfers electric supply power to 59V DC power supply; charges the battery and communicates with the charger management main board; and transfers 59V DC power to the charger main board.

3-pin Adapter: Connects the three-hole plug and 4850 module.



LED Button Board: Connects to the switch, transfers the switch, and manages the main board;
Has 7 LEDs which represent the charging status of each port, alarm, fast charging, and slow charge display.

Battery Charging Port



Connects the red cable to the left small screw hole (positive hole). There is “+” in the port. Connects the blue cable to the right large screw hole (negative hole). There is “-” in the port. The signal cable is connected to the middle port.

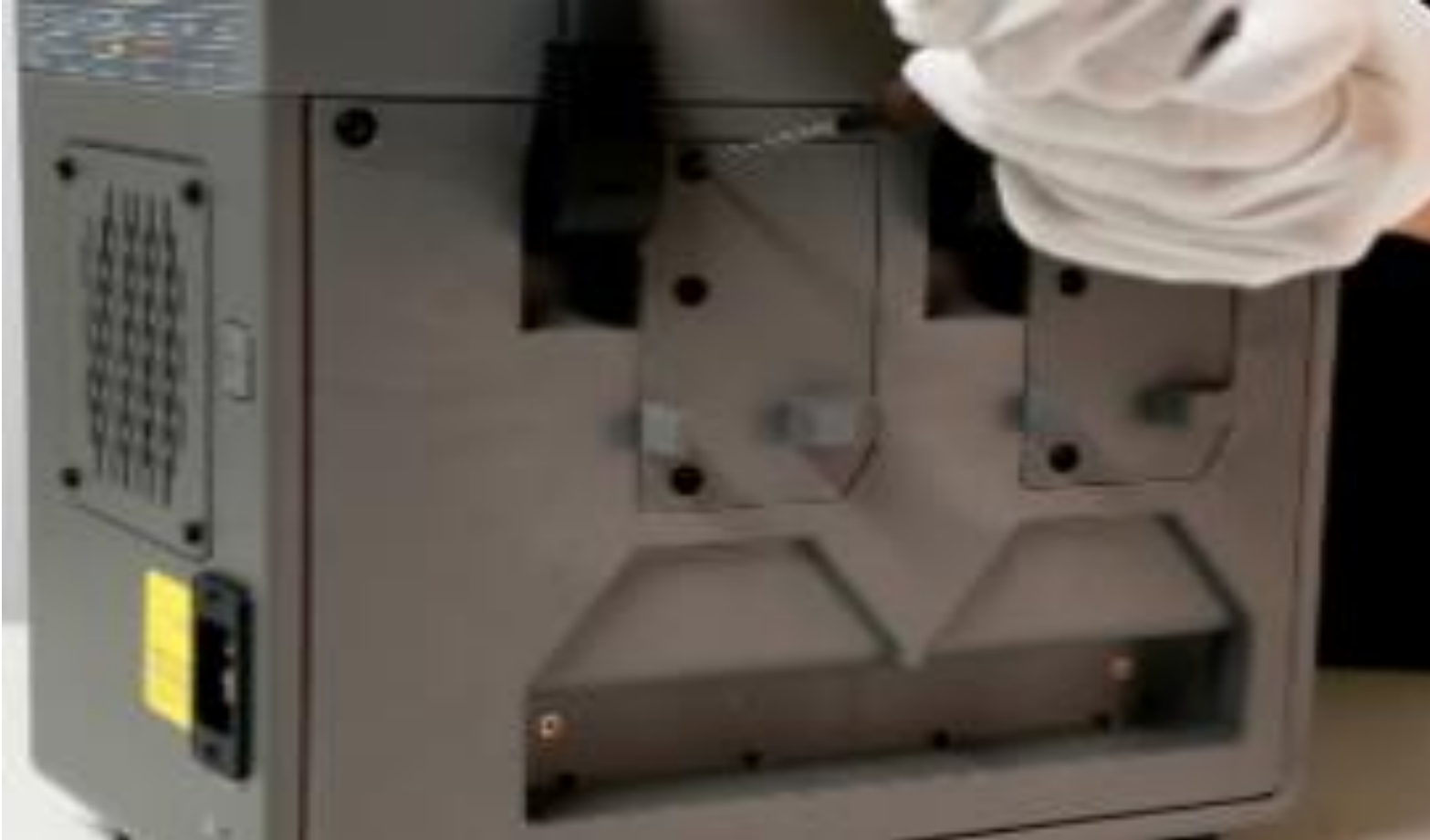
Disassembling and Assembling the 4-Channel Battery Charger

Disassembling and Assembling the 4-Channel Battery Charger



Remove the four charger plugs.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the six screws securing the two charger side boards C on the charger side board A module, then remove the charger side boards C.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the battery charging cables. There is a cable clamp on the communication cable, so it cannot be removed with too much force. Note that the red cable is attached to the left small screw hole, and the blue cable is on the larger right hole. The cables must be attached to the correct positive and negative poles, and the screws on the left and right are different. Torque for installation: 5.0-5.5 kgf.cm

Disassembling and Assembling the 4-Channel Battery Charger



Remove the four screws on the charger bottom pads and then remove the pads.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the four screws
securing the charger upper
cover module.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the upper cover and disconnect the charger button board's communication cable.
Slowly remove the upper cover to avoid damaging the communication cable.

Disassembling and Assembling the 4-Channel Battery Charger



Disconnect the switch cable and remove the switch with the pliers.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the four screws securing the LED board, then remove the LED board and shielding foam.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the charger silicone rubber stopper and button cap.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the two screws' shielding pieces on the handle.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the two screws and nuts securing the handle.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the four screws securing the charger fixing plate, then remove the plate.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the two charger side board A modules.

When installing, arrange the input cables to prevent them from being crushed during installation.

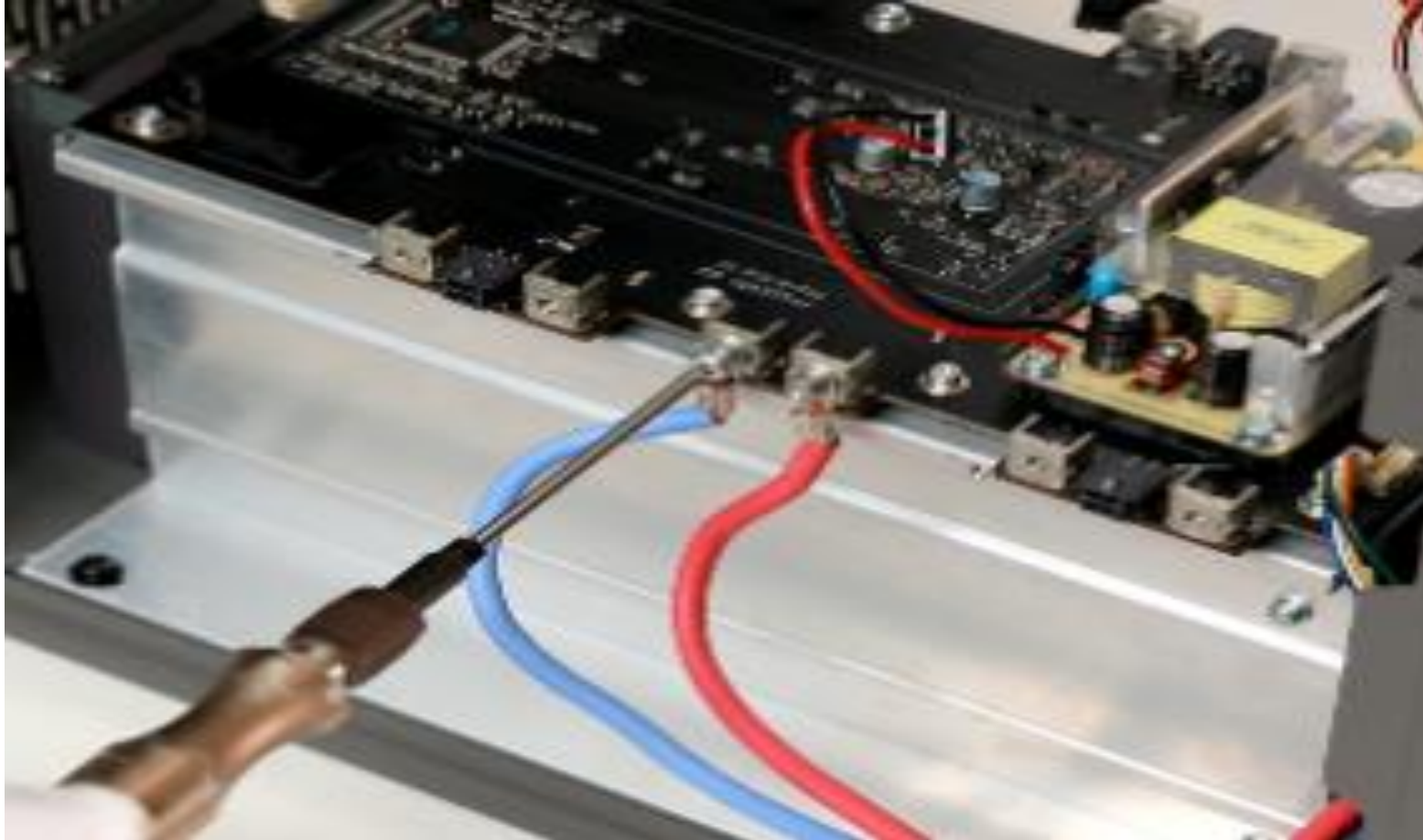
Disassembling and Assembling the 4-Channel Battery Charger



Disconnect the adaptive board's communication cable, fan's cable, four core cable connecting the USB Type-C to main board, AC power cable, auxiliary power board cable, and button board's communication cable.

Do not pull out the cable with the buckle. Note that there is a locating pole of the auxiliary power cable. Make sure that the fire wire and zero line are attached correctly.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the screws securing the AC power cable (positive red cable) and AC power cable (negative blue cable).

During installation, ensure that the red cable is attached to the small screw hole (positive pole) while the blue cable is attached to the large screw hole (negative pole).

Torque: 5.0-5.5 kgf.cm

Disassembling and Assembling the 4-Channel Battery Charger



Remove the four screws securing the auxiliary power board, then remove the board. Remove the charger auxiliary power vacuum thermoform cover and main board anti-dust vacuum thermoform cover.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the eight screws securing the charger management main board PCBA (with firmware), then remove the PCBA.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the eight cooling pads.
If the cooling pad is replaced,
thermal grease must be
applied.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the four screws securing the 4850 power board module and GND, then remove the module.

Disassembling and Assembling the 4-Channel Battery Charger



Disconnect the four core cable connecting USB Type-C and main board, remove the two screws securing USB update adaptive board, then remove the adaptive board.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the two screws
securing the AC power plug,
then remove the plug.

**Pay attention to the installation
direction as shown.**

Disassembling and Assembling the 4-Channel Battery Charger



Remove the four screws securing the fan's metal cover, then remove the cover.

If only the fan is damaged, there is no need to remove the upper cover. Directly remove or install the fan.

Disassembling and Assembling the 4-Channel Battery Charger



Remove the four screws securing the fan and then remove the fan and foam.

Pay attention to the installation direction. Make sure that the sticker with the silk print faces outward, and that the cable is near the yellow sticker.

Disassembling and Assembling the 4-Channel Battery Charger

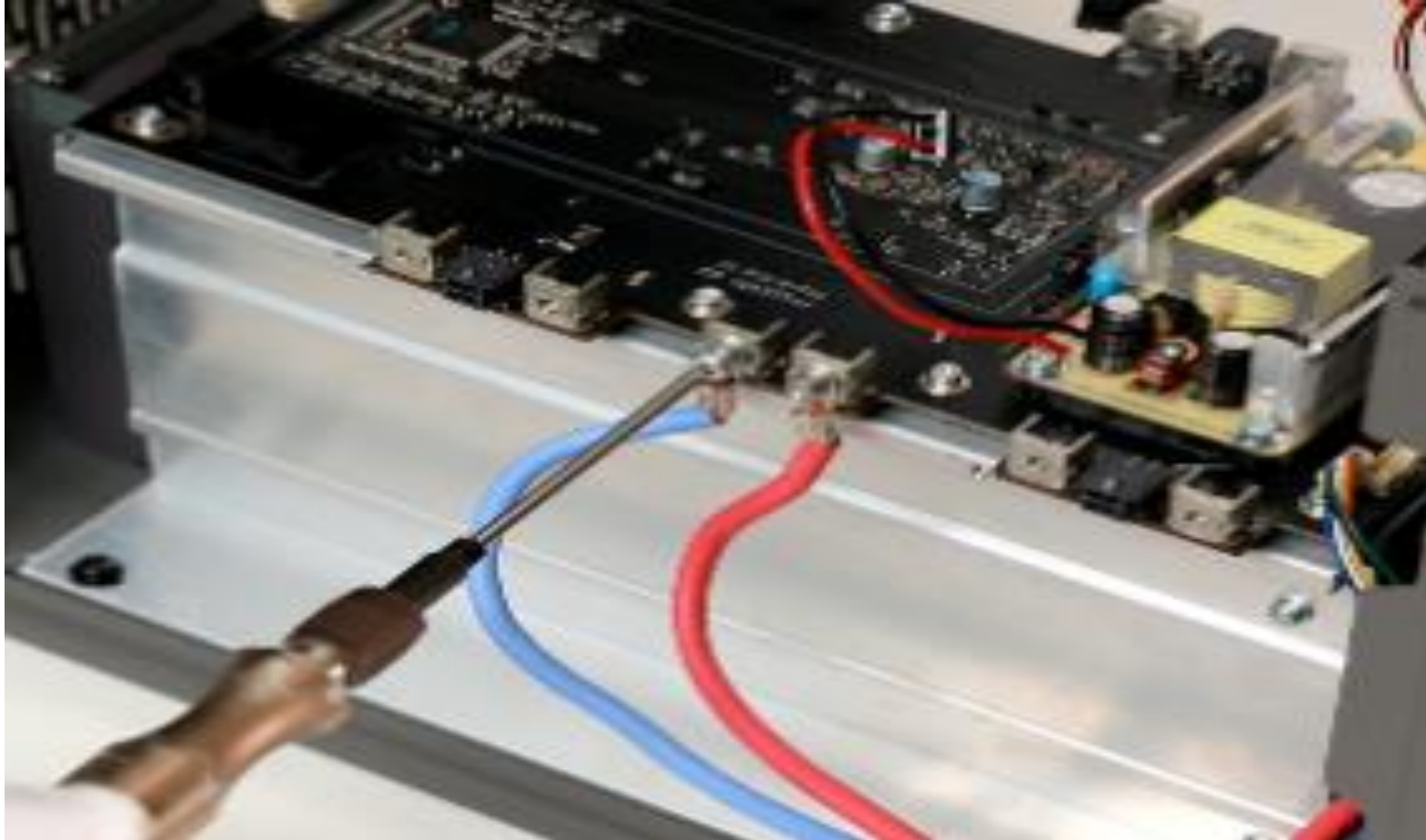


Remove the whole 4850 module.

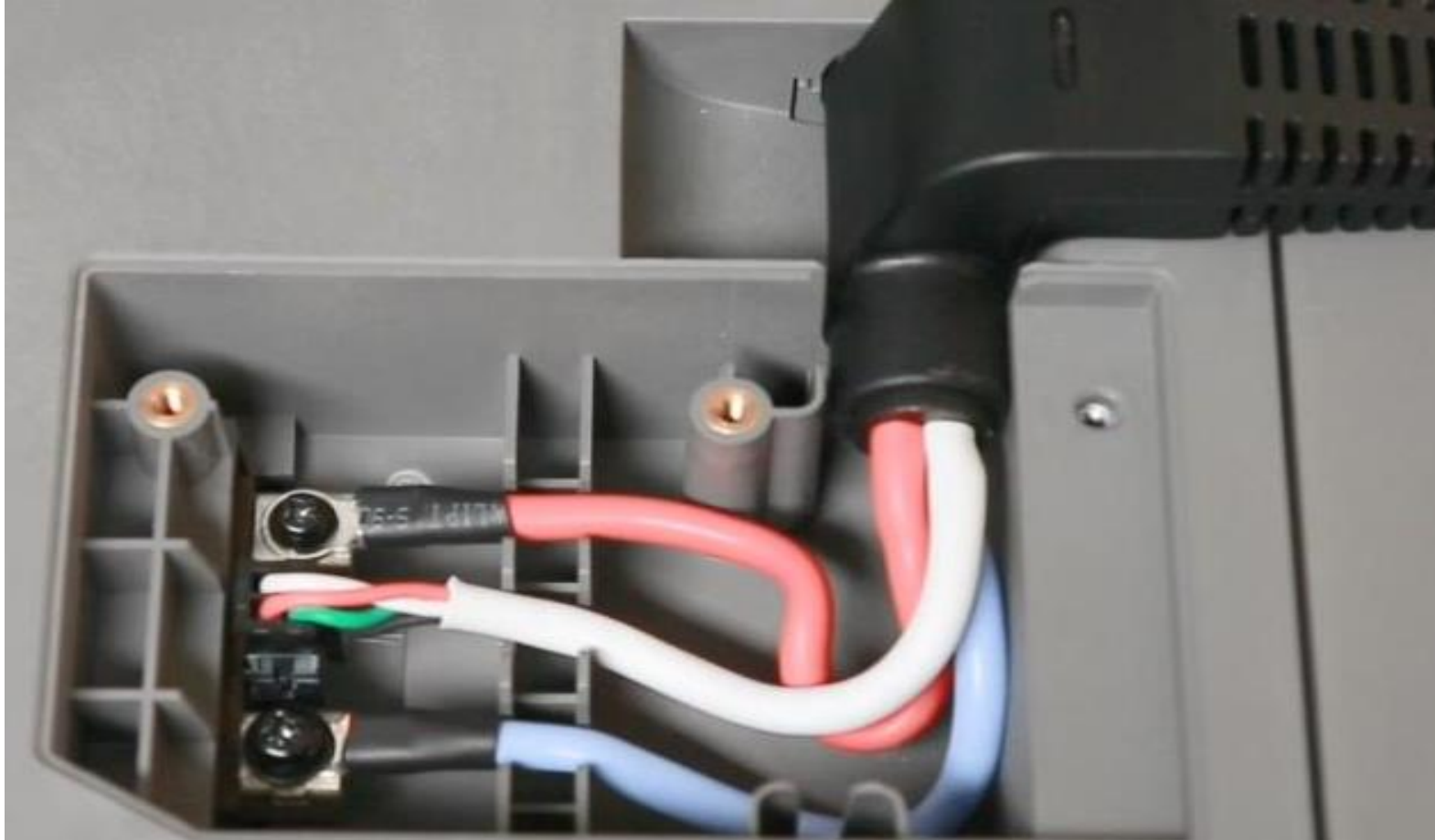
Disconnect the AC power input cables (negative and positive), AC power input cable GND, AC power output blue cable (negative), AC power output red cable (positive), and six core cable connecting the 4850 module adaptive board and main board.

Ensure that the fire wire, zero line, and earth wire are attached correctly. The red and blue cables cannot be connected incorrectly.

Torque: 5.5 ± 0.5 kgf.cm



Ensure that the torque is 5.0-5.5 Kgf.cm when connecting the red and blue power cables.



Ensure that the torque is 5.0-5.5Kgf.cm when connecting the battery charging cable.

Troubleshooting



Issue	Cause Analysis	Solution
The charger does not power on (the LED does not light up and the fan does not rotate).	The auxiliary power is damaged, so there is no 12 V output for the charger management main board.	Replace the auxiliary power.
	The charger management main board is damaged.	Replace the charger management main board.
	The switch is damaged.	Replace the switch.
	The cable connecting the button board and the charger management main board is damaged.	Replace the corresponding cable.
	The AC power cable is damaged, or the negative and positive cables are loose.	Replace or reinsert the AC power cable, negative cable, and positive cable.
The charger can be powered on, but the ports cannot be used to charge.	The 4850 power module is damaged.	Replace the 4850 power module.
	The charger management main board is damaged.	Replace the charger management main board.
One port cannot be used to charge.	The corresponding charging cable is not connected well or damaged.	Perform a visual inspection and replace the charging cable.
	The corresponding charging cable is not connected well or damaged.	Perform a visual inspection and replace the charging plug.
	The charger management main board is damaged.	Replace the charger management main board.
The power button LED lights up, but the fan does not rotate.	The fan is damaged.	Replace the fan.
	The corresponding charging cable is not connected well or damaged.	Replace the charger management main board.

Troubleshooting



Issue	Cause Analysis	Solution
Insert the battery and the corresponding port's LED blinks yellow continuously.	The corresponding charging cable is damaged or not connected well.	Perform a visual inspection and replace the charging cable.
	The corresponding charging plug is damaged or not connected well.	Perform a visual inspection and replace the charging plug.
Insert the battery and the corresponding port's LED blinks yellow twice or four times.	There is an error with the battery.	Replace the battery.
Insert the battery and the corresponding port's LED blinks continuously (after the battery is inserted for about 10 seconds).	The battery is not inserted well or a non-official battery is used.	Replace the battery.
Insert the battery and the corresponding port's LED blinks yellow twice.	The port is short-circuited.	Check whether the charging plug or charging cable is short-circuited.
Insert the battery and the corresponding port's LED does not light up and the warning indicator is solid yellow.	The 4850 power module is damaged.	Replace the 4850 power module.
	The charger management main board is damaged.	Replace the charger management main board is damaged.
Insert the battery and the corresponding port's LED does not light up and the warning indicator blinks twice.	The fan of the 4850 power module is damaged.	Replace the fan and 4850 power module.
Insert the battery and the corresponding port's LED does not light up and the warning indicator blinks three times.	The charger's fan is damaged.	Replace the charger's fan.
Insert the battery and the corresponding port's LED does not light up, the warning indicator blinks once, and the charger makes "DDD" sounds.	The fan of the 4850 power module is damaged.	Replace the 4850 power module.
Insert the battery and the corresponding port's LED does not light up, the warning indicator lights up, and the charger makes "DD.DD.DD" sounds.	The charger management main board is damaged.	Replace the charger management main board is damaged.
The USB port cannot be used to connect and update.	The USB adaptive board is damaged.	Replace the USB adaptive board.
	The charger management main board is damaged.	Replace the charger management main board is damaged.
The fast charging and slow charging modes cannot be switched.	The LED board is damaged.	Replace the LED board.
	The charger management main board is damaged.	Replace the charger management main board is damaged.

Troubleshooting



Issue	Cause Analysis	Solution
Charger One Channel Malfunction	1. Battery malfunction 2. Charger charging plug malfunction 3. Charger cable malfunction 4. Charger inner error	Replace the corresponding module or main board.
Unable to power on	1. Charger AC cable malfunction 2. Auxiliary module malfunction 3. Button communication cable malfunction 4. Button board malfunction	Replace the corresponding module.
LED does not light up after being powered on	1. Auxiliary module malfunction 2. Button cable malfunction 3. Button board malfunction	Replace the corresponding module.
Alarm occurs after being powered on	1. Power cable malfunction 2. Main board cable loose or malfunctioning 3. Main board malfunction	Replace the corresponding module.
Charging normally, but the actual battery level cannot be increased	1. Battery malfunction 2. Charger cable and charging plug malfunction 3. Positive and negative cables malfunction (red and blue cable)	Replace the corresponding module.
Unable to switch between Charging Modes	1. Button communication cable malfunction 2. Button board malfunction	Replace the corresponding module.
After unactivated 2600W 4-channel charger is powered on, the LEDs 1234 do not light up and the warning indicator lights up. Press the Mode Button and the charging mode can be chosen, but the battery cannot be charged.	The charger is not activated.	Use DJI Assistant 2 to activate the charger.

**THANK
YOU**

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